

SafeTransit

SafeTransit is a comprehensive safety-first navigation mobile application designed for women commuters in Philippine cities. Built with React Native, Expo SDK 54, and NativeWind (Tailwind CSS), it empowers women travelers with real-time safety information, community-powered alerts, and intelligent route planning to create a complete journey protection network.

Features

Safety and Protection

- Background Protection that monitors location continuously, even when app is closed, and alerts on entering danger zones
- Silent Panic Button for discreet emergency alerts sent to helpers and emergency contacts
- Quick Exit Screen that instantly disguises the app as a weather forecast for privacy protection
- Danger Zone Alerts with real-time notifications when approaching high-risk areas
- Safety Heatmap visual overlay showing area safety levels based on community data

Navigation and Routing

- Safe Route Planning with multi-modal options (Walking, Driving, Transit) and real-time safety assessments
- Color-Coded Routes showing danger zones, safe segments, and overall route safety score
- Turn-by-Turn Navigation with step-by-step guidance, distance, and direction indicators
- Route Safety Analysis based on nearby safety tips and danger zones
- Hybrid Routing Engine using OpenRouteService for pedestrian routes and LocationIQ for driving routes

Community Map Tips

- Location-specific, admin-verified safety alerts with categories (Poor Lighting, Harassment, Transit Issues, Safe Havens, Construction)
- Tip markers color-coded by severity (High/Critical, Medium, Low)
- Intelligent clustering for better map readability
- Filter by category, radius, and time relevance

- Auto-expanding tip detail cards with full information
- Admin verification system for tip quality

Community Forum

- Discussion-based posts with upvoting and downvoting system
- Threaded comments with nested replies
- Post flairs (Safety Alert, Discussion, Question, etc.)
- Location tags for area-specific discussions
- Photo attachments via UploadThing
- Report and moderation system

Family Location Sharing

- Real-time location tracking with trusted family members
- Live location indicators on map
- Quick navigation to family member locations
- Privacy controls and permission management

Authentication and Security

- JWT-based Authentication with secure token storage
- Email and Password Login with input validation
- Google Sign-In integration
- Token refresh mechanism for persistent sessions
- Background location permissions with user consent
- Secure API communication with bearer tokens

User Experience

- Modern UI with smooth animations and glassmorphic design
- Fully responsive layout for all Android screen sizes
- Intuitive navigation with Expo Router (file-based routing)
- Interactive map with clustering and custom markers

- Haptic feedback for important interactions
- Compatible with Android 11-16 (SDK 30-35)

Tech Stack

Frontend

- React Native 0.81.5 - Cross-platform mobile framework
- Expo SDK 54 - Development and build tooling
- Expo Router 6.0.21 - File-based navigation with typed routes
- TypeScript 5.9 - Full type safety
- NativeWind 4.0 - Tailwind CSS for React Native
- React Native Reanimated 4.1 - Smooth 60fps animations
- React Native Gesture Handler - Advanced gesture support
- React Native Maps 1.20 - Google Maps integration with clustering
- Lucide React Native - Icon library

Backend and Services

- Vercel - Serverless functions for API endpoints
- Neon PostgreSQL - Serverless database with HTTP/WebSocket driver
- UploadThing - Image upload service
- JWT - Token-based authentication
- AsyncStorage - Local data persistence

External APIs and Services

- Google Maps SDK - Map rendering and geocoding
- LocationIQ - Driving route planning and geocoding
- OpenRouteService - Pedestrian and cycling route planning
- Nominatim - Reverse geocoding (OpenStreetMap)
- Overpass API - OpenStreetMap data queries

Architecture

- Service Layer Architecture - Separation of business logic
- Repository Pattern - Data access abstraction
- File-Based Routing - Expo Router conventions
- Component-Driven Development - Reusable UI building blocks

Getting Started

Prerequisites

- Node.js 20+
- npm package manager
- Android Studio with Android SDK installed
- Physical Android device with USB debugging enabled (recommended over emulator)
- Google Maps API key (free tier available - \$200/month credit)

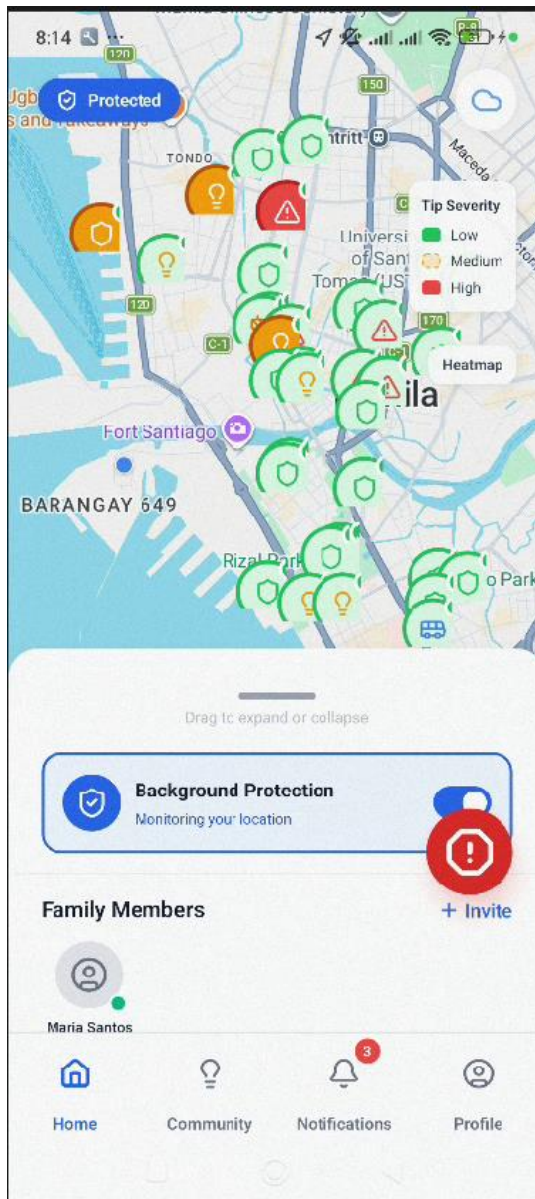
Installation

1. Clone the repository `git clone <your-repo-url> cd SafeTransit`
2. Install Dependencies
 - Run `npm install` to install all project dependencies
 - Patch packages will run automatically via postinstall hook
3. Configure Environment
 - Set up Android SDK path configuration
 - Add Google Maps API key to environment variables
 - Configure Neon PostgreSQL connection string
 - Configure UploadThing credentials
4. Build and Install (First Time)
 - Run `npx expo run:android`
 - This builds and installs the development build on your device
 - You only need to do this once per device

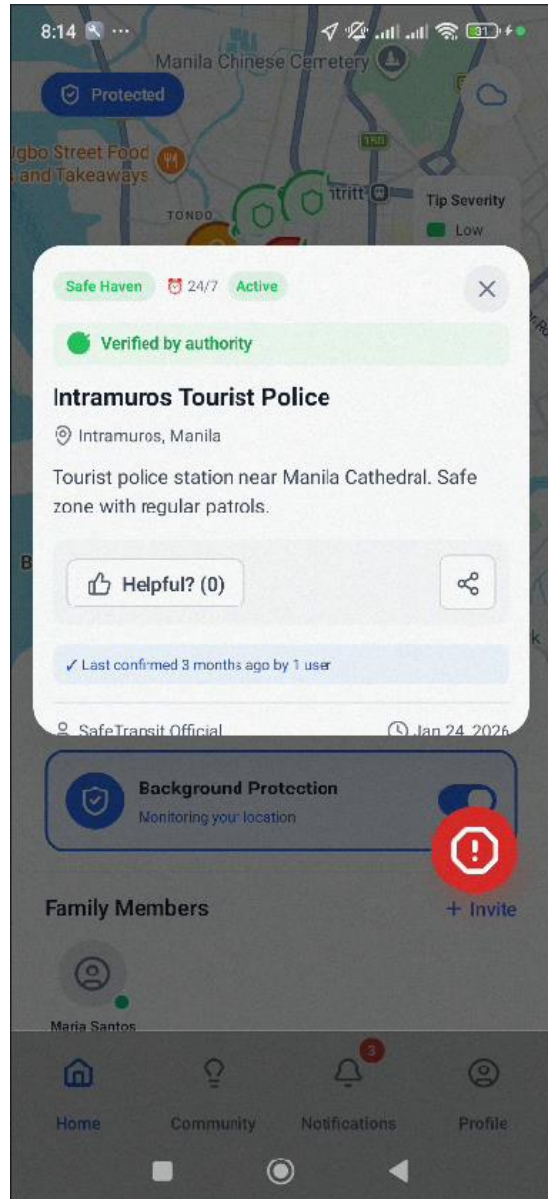
5. Start Development Server

- Run npm start
- Press a to open on Android, r to reload

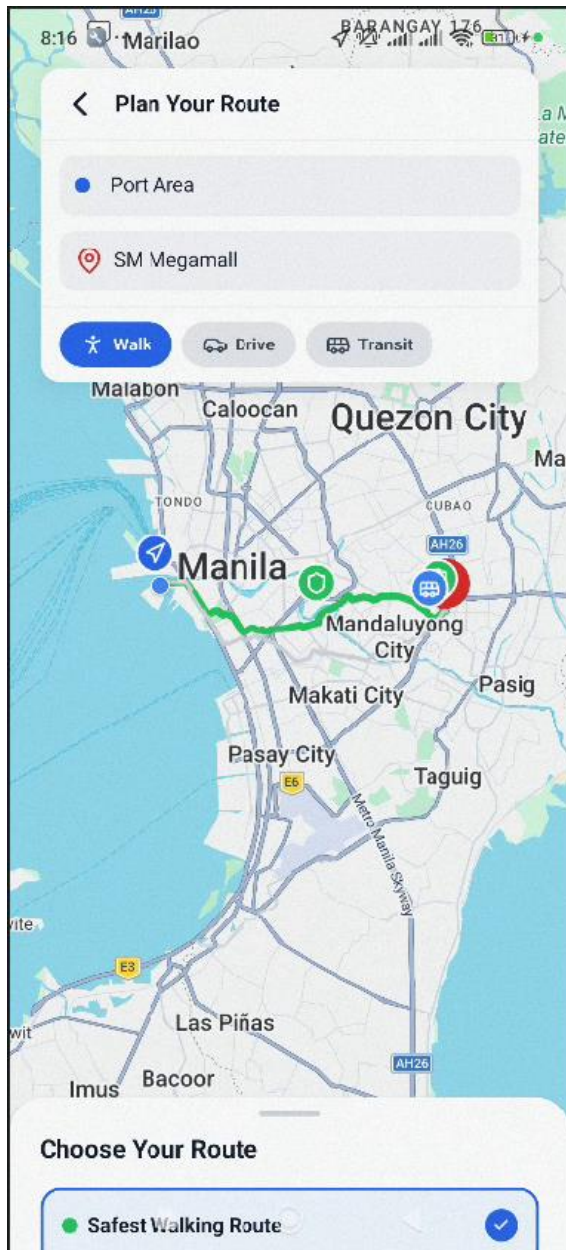
Page Overview



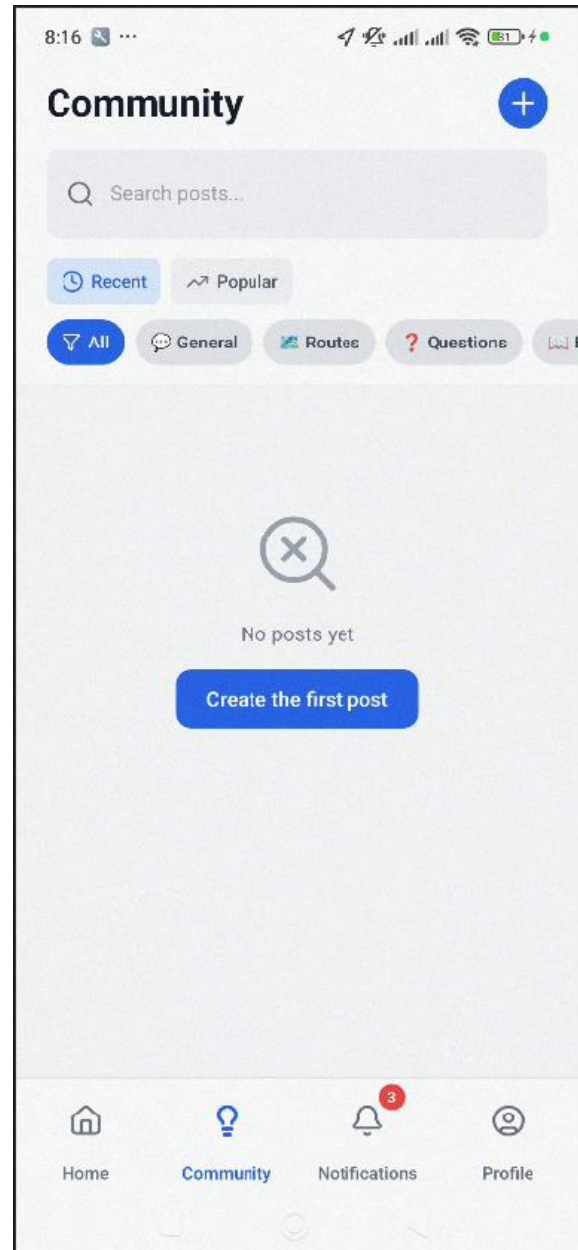
HOME



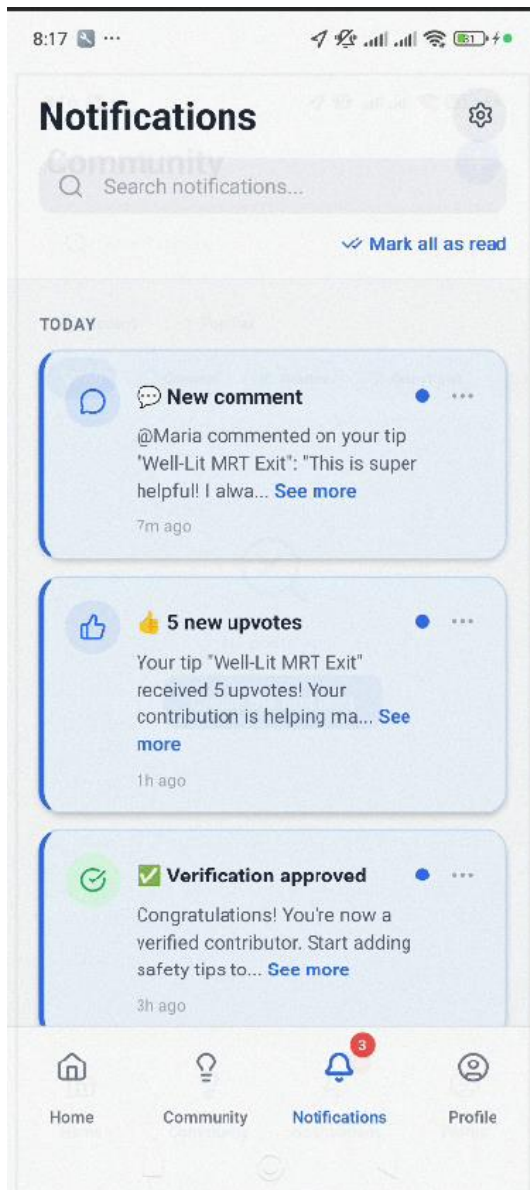
PIN



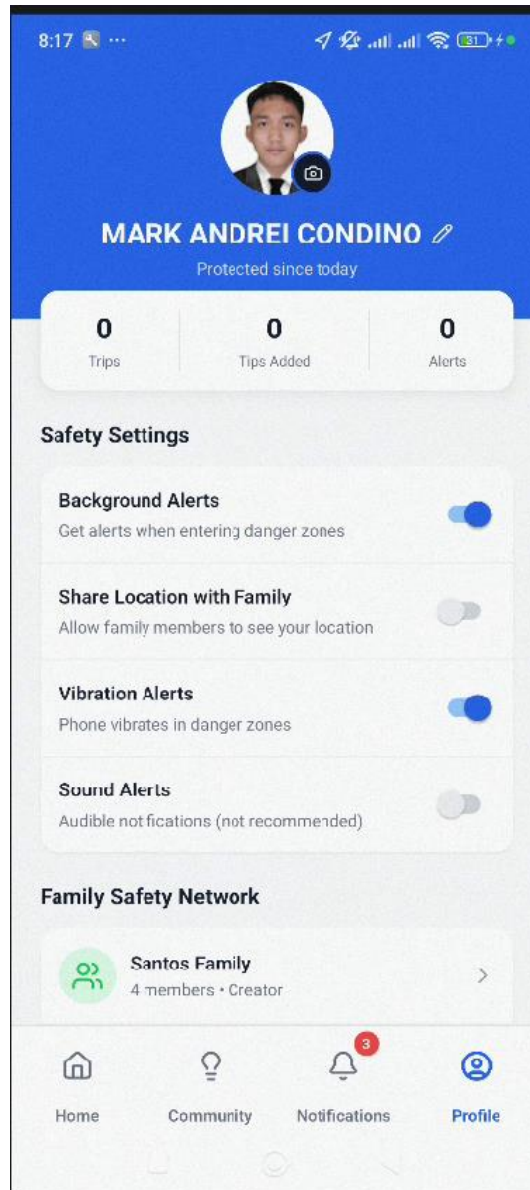
ROUTE



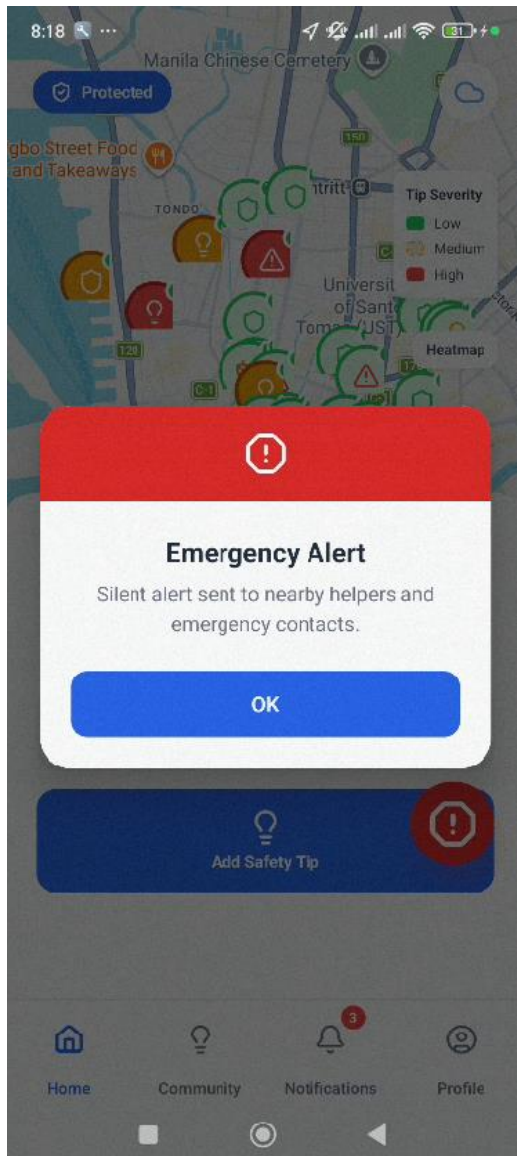
FORUM



NOTIF



PROFILE



SOS